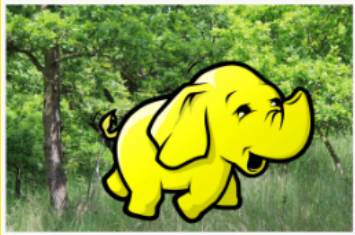


The Hadoop Ecosystem



Frank Kane

Hadoop Overview

What is Hadoop?

The open source software platform for distributed storage and processing of large data sets across commodity hardware. It is designed to scale up to large numbers of machines, and to support high throughput, streaming data access.

Why Hadoop?



- Scalable architecture: Hadoop can scale from a single node to thousands of nodes.
- Fault tolerance: Hadoop can handle hardware failures without losing data.
- High throughput: Hadoop can process large volumes of data quickly.
- Cost-effective: Hadoop can run on commodity hardware.

Hadoop History



Hadoop was created by Doug Cutting and Mike Jones at Yahoo! in 2005. It was inspired by Google's MapReduce and GFS. Hadoop is now a part of the Apache Software Foundation.

World of Hadoop

Query Engines



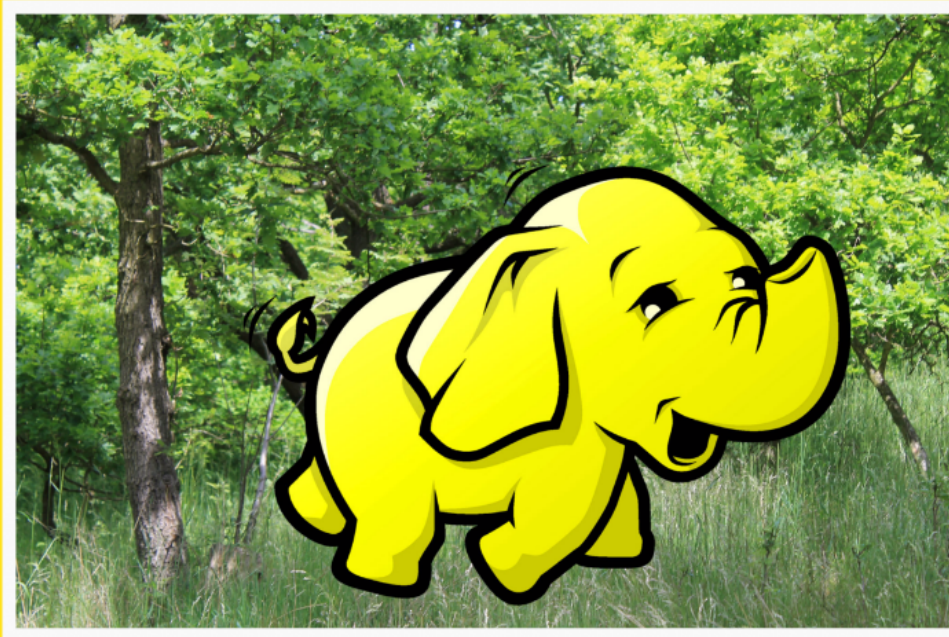
Core Hadoop Ecosystem



External Data Storage



The Hadoop Ecosystem



Frank Kane

Hadoop Overview

What is Hadoop?

"an open source **software platform** for **distributed storage** and **distributed processing** of **very large data sets** on **computer clusters** built from commodity hardware" - *Hortonworks*

Why Hadoop?



- Data's too darn big - terabytes per day
- Vertical scaling doesn't cut it
 - Disk seek times
 - Hardware failures
 - Processing times
- Horizontal scaling is linear
- Hadoop: It's not just for batch processing anymore

Hadoop History



- Google published GFS and MapReduce papers in 2003-2004
- Yahoo! was building "Nutch" an open source web search engine at the same time
- Hadoop was primarily driven by Doug Cutting and Tom White in 2006
- It's been evolving ever since...

What is Hadoop?

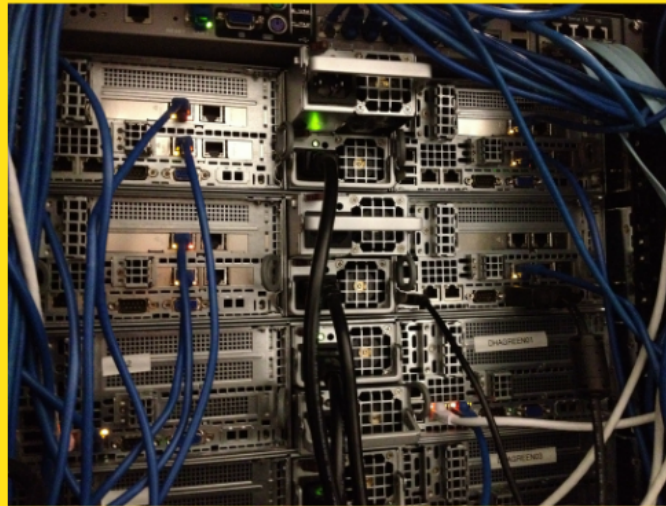
"an open source **software platform** for **distributed storage** and **distributed processing** of **very large data sets** on **computer clusters** built from commodity hardware" - *Hortonworks*

Hadoop History



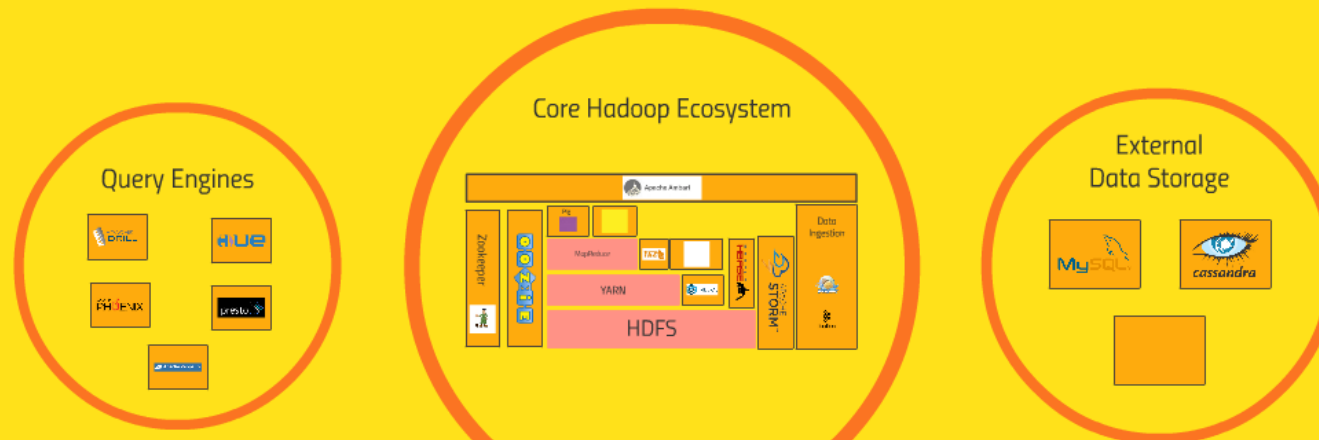
- Google published GFS and MapReduce papers in 2003-2004
- Yahoo! was building "Nutch," an open source web search engine at the same time
- Hadoop was primarily driven by Doug Cutting and Tom White in 2006
- It's been evolving ever since...

Why Hadoop?

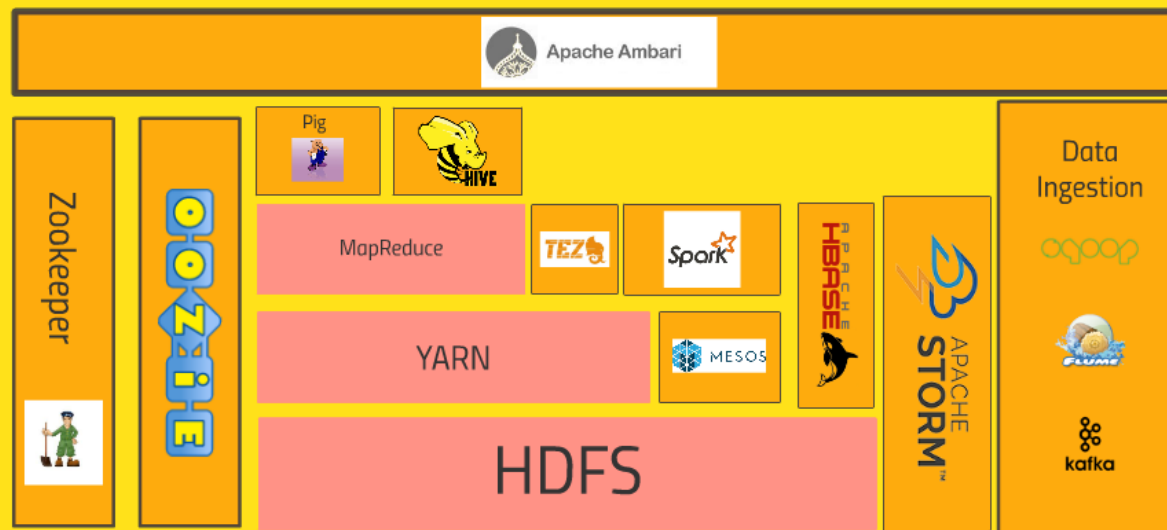


- Data's too darn big - terabytes per day
- Vertical scaling doesn't cut it
 - Disk seek times
 - Hardware failures
 - Processing times
- Horizontal scaling is linear
- Hadoop: It's not just for batch processing anymore

World of Hadoop



Core Hadoop Ecosystem



MapReduce



YARN



HDFS

MapReduce



YARN

HDFS



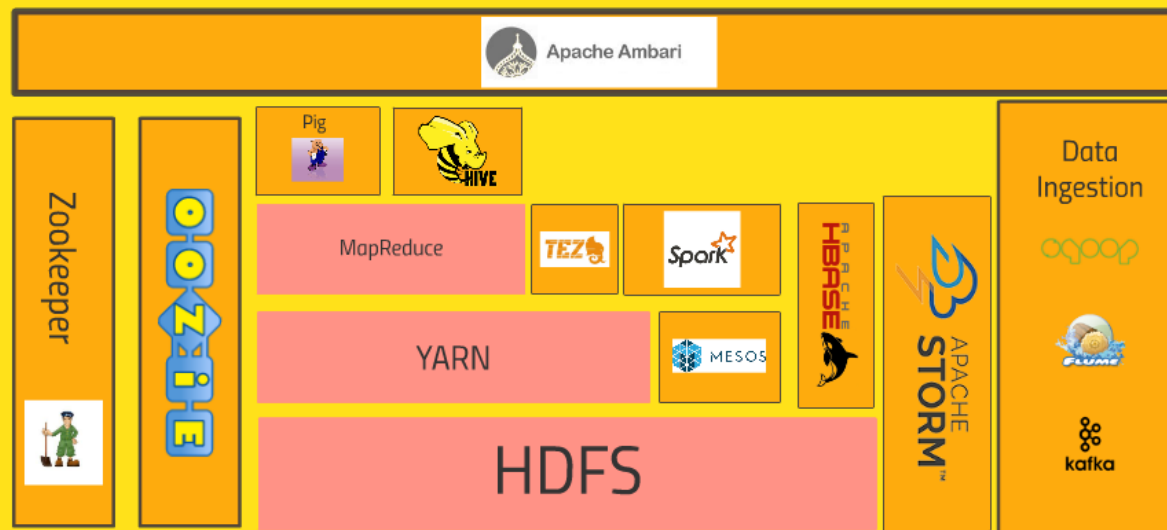
MapReduce

VADN

Pig



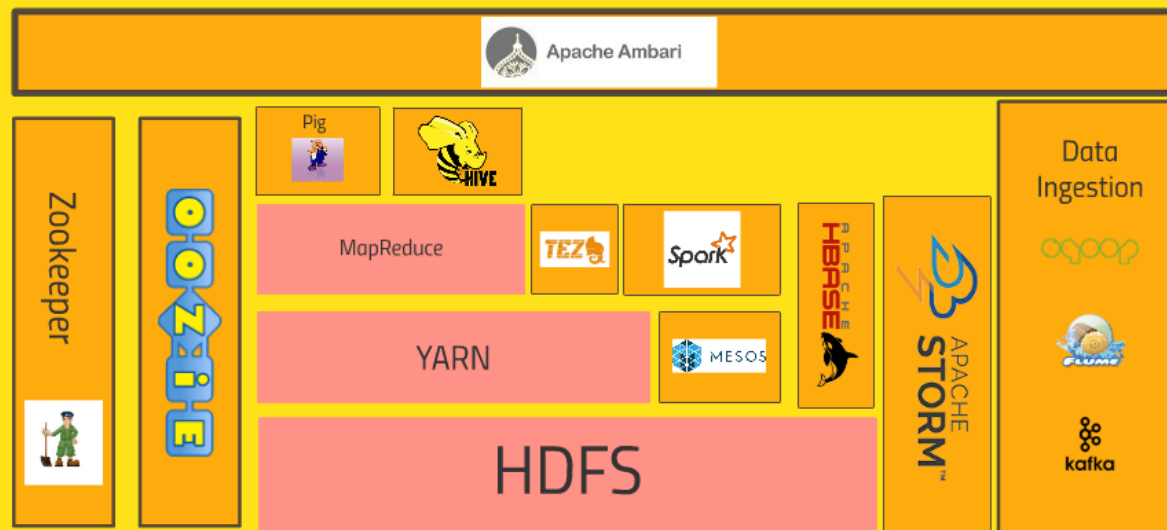
Core Hadoop Ecosystem





HIVE

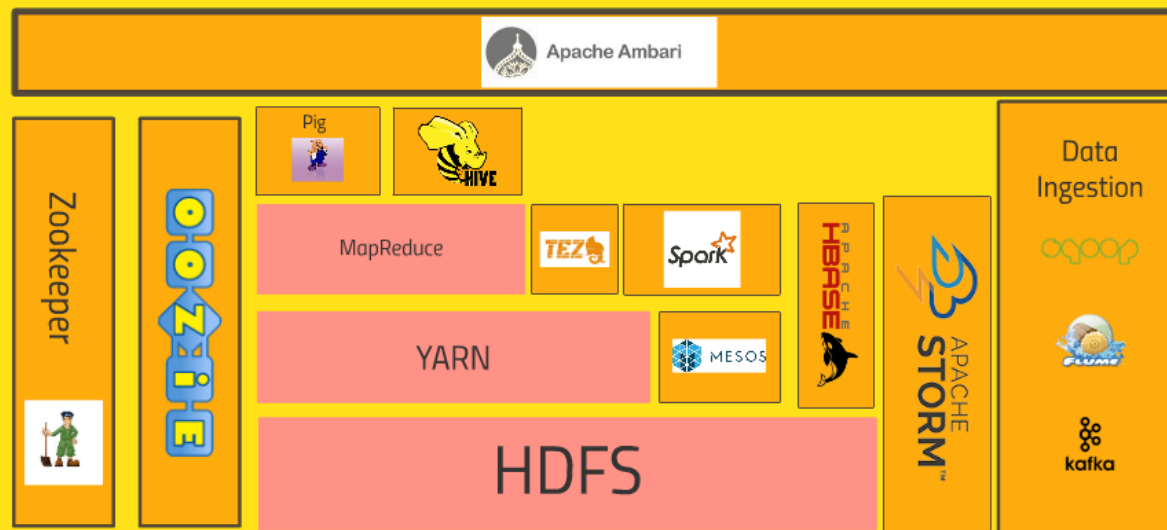
Core Hadoop Ecosystem





Apache Ambari

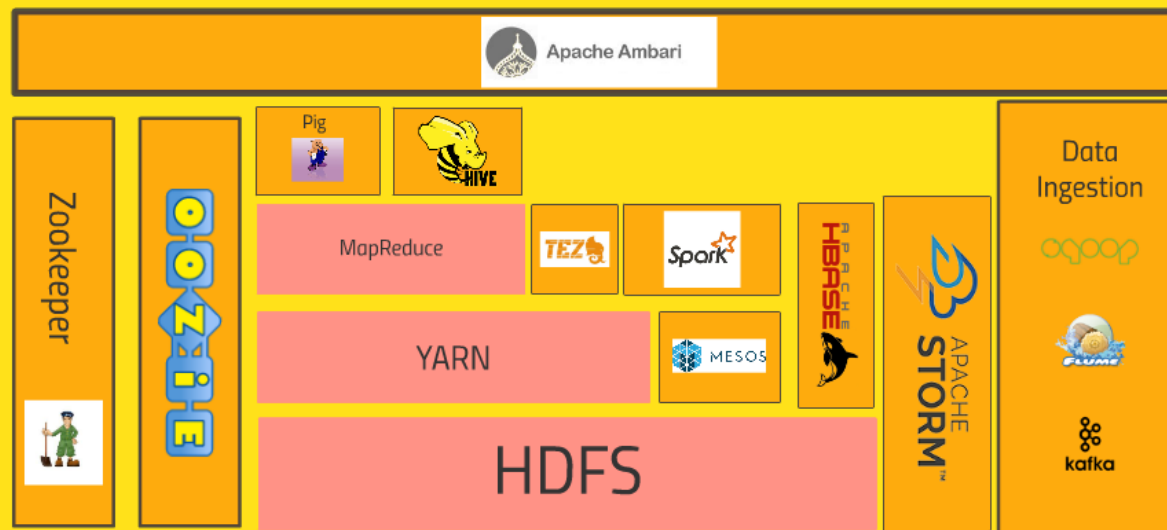
Core Hadoop Ecosystem





MESOS

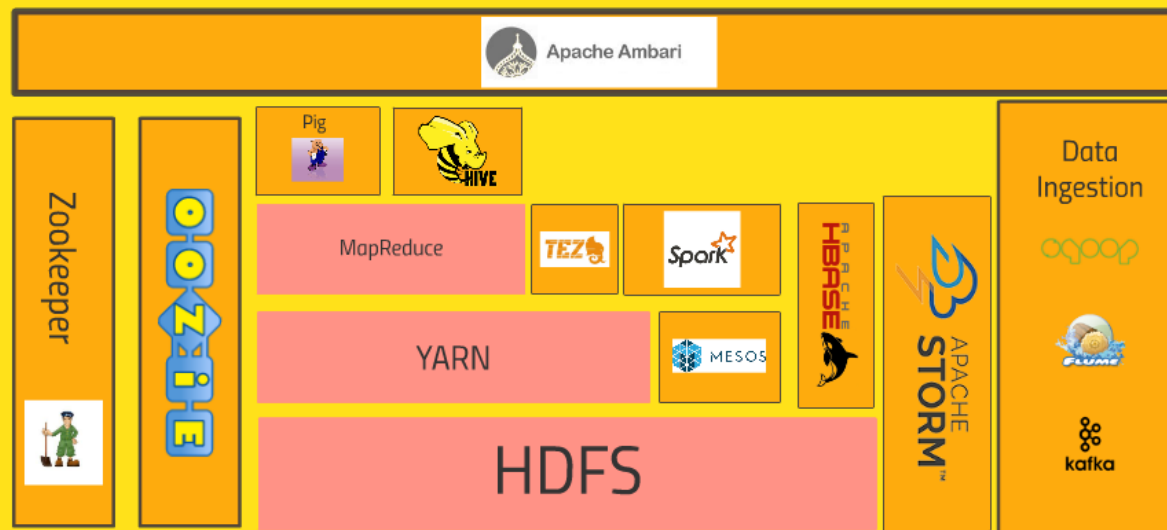
Core Hadoop Ecosystem



Spark

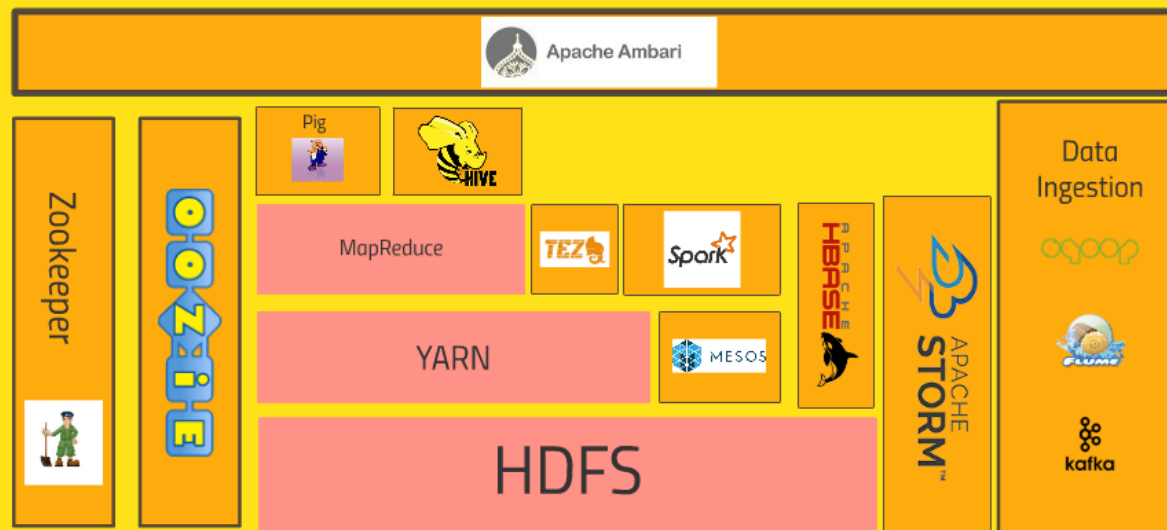
An orange outline of a five-pointed star is positioned behind the letter 'k' in the word 'Spark'. The star is slightly offset to the right and upwards, creating a layered effect. The word 'Spark' is written in a bold, dark grey, sans-serif font. The entire logo is centered on a white background, flanked by two vertical orange bars on the left and right sides.

Core Hadoop Ecosystem





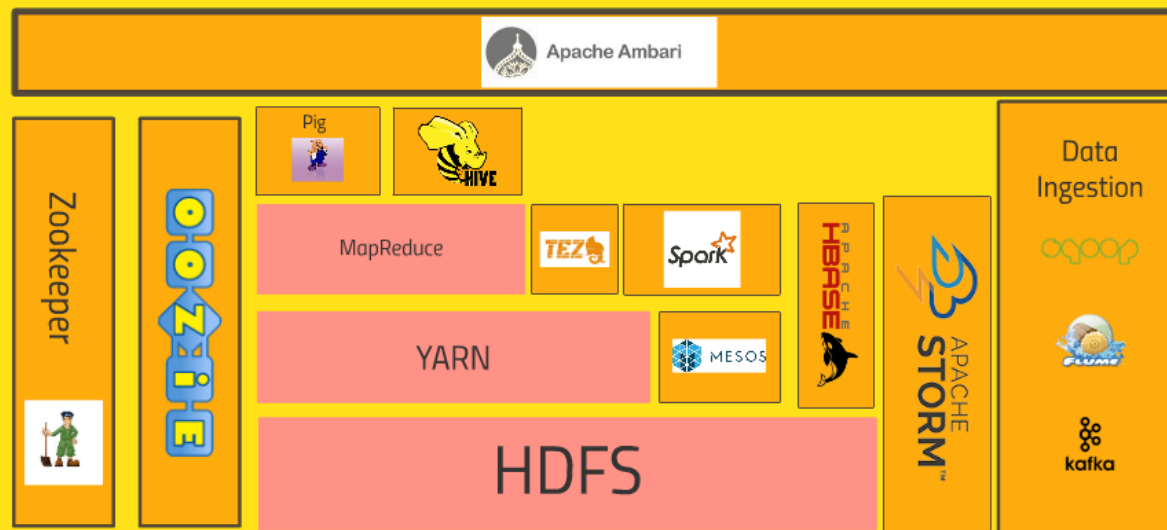
Core Hadoop Ecosystem



A P A C H E
H B A S E



Core Hadoop Ecosystem



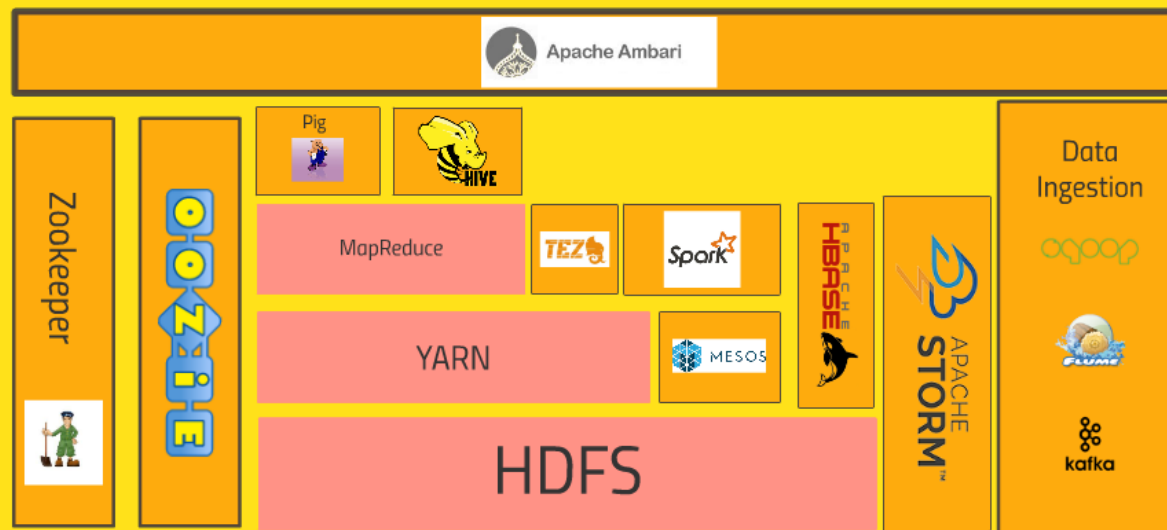


APACHE
STORMTM

P A C H E



Core Hadoop Ecosystem



Zookeeper



Pig



MapReduce

YARN

Pig



M

oozie

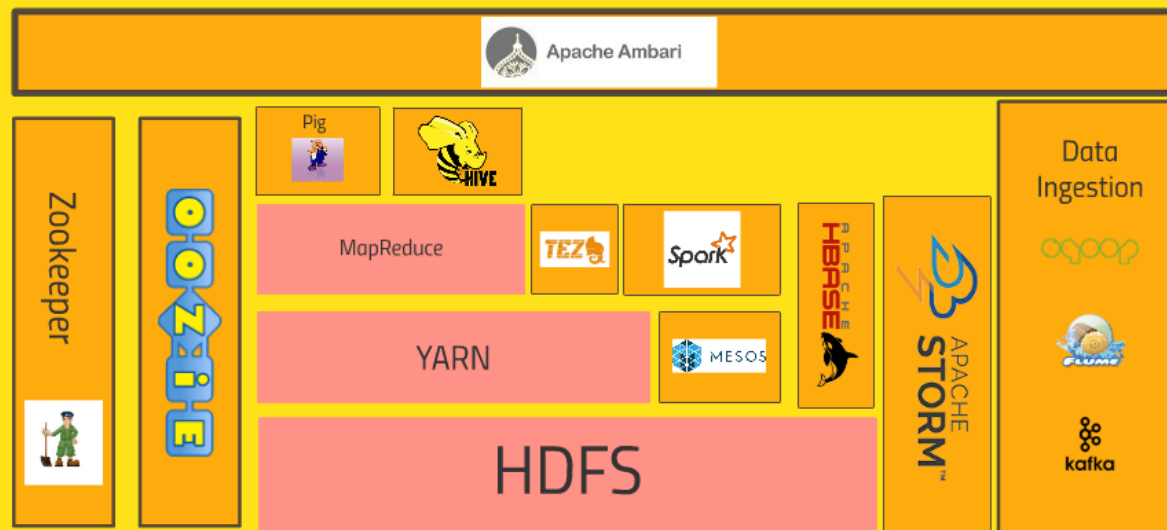
Zookeeper



Data Ingestion



Core Hadoop Ecosystem



External Data Storage





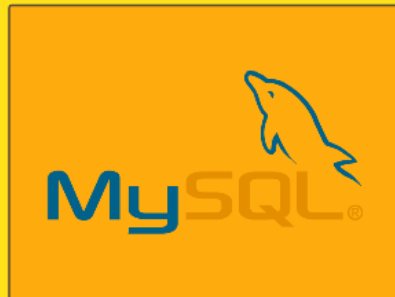


cassandra



mongoDB®

External Data Storage



Query Engines





APACHE
DRILL

hue

APACHE
PHOENIX

presto





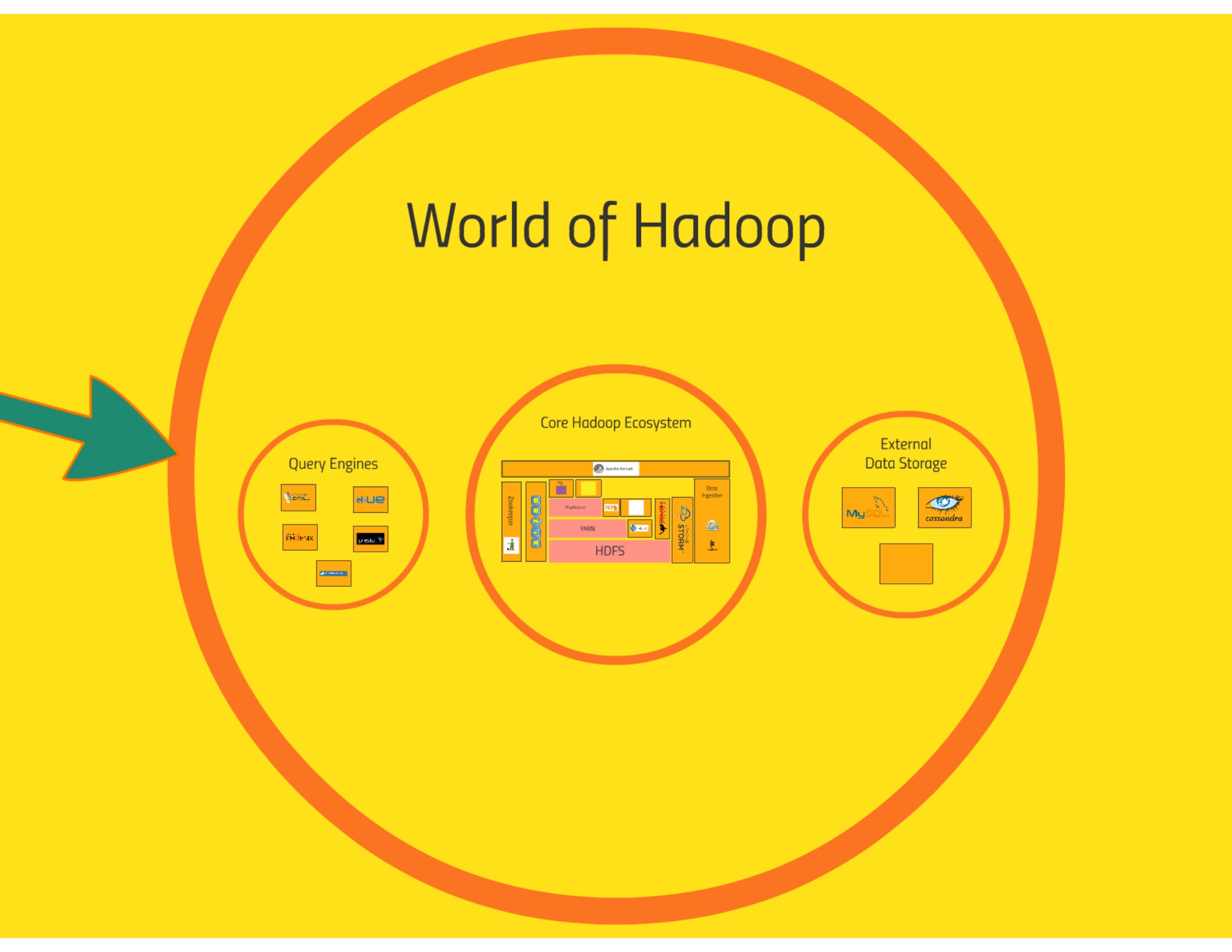
Apache Zeppelin

World of Hadoop

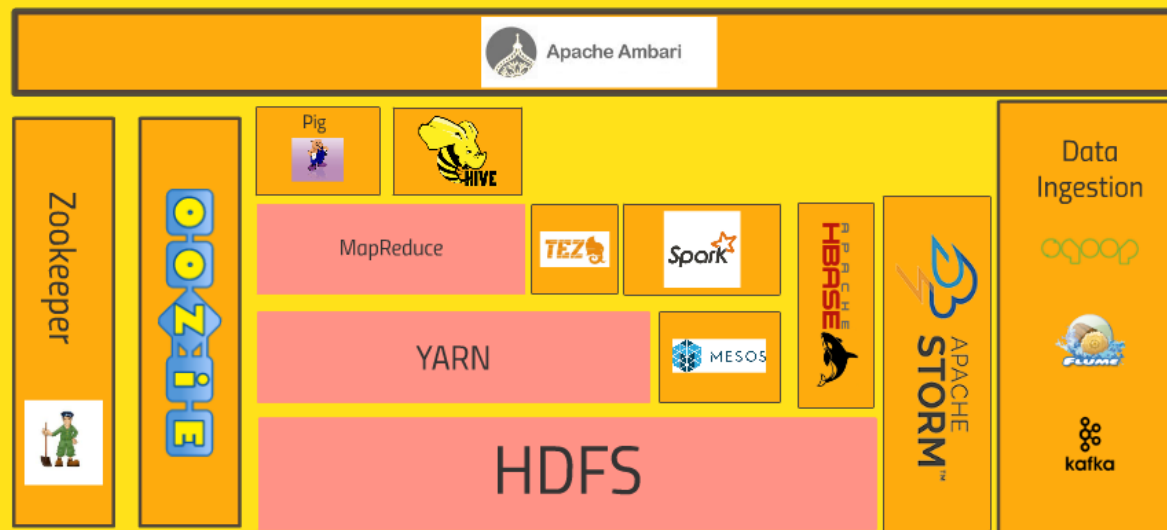
Query Engines

Core Hadoop Ecosystem

External Data Storage



Core Hadoop Ecosystem



External Data Storage



Query Engines

